treated Hooke as an equal, and they worked as partners on the design and build of the first air pump during their investigations into vacuums. These led them to the discovery of Boyle's law on gas compression, yet Boyle himself acknowledged Hooke's contribution.

While working with Boyle, Hooke also carried out many experiments of his own. In 1657, he significantly improved the pendulum clock by inventing the anchor escapement—a vital mechanism that maintains the pendulum's regularity. He also studied the elasticity of springs and applied his findings to the art of clockmaking, inventing the balance-spring for pocket watches, which enabled accurate timekeeping. In 1660, he formulated his law of elasticity, later known as Hooke's law, which states that the tension in a spring increases in direct proportion to the force applied to it.

Curator of experiments

Hooke became part of a circle of like-minded intellectuals who engaged in regular meetings aimed at scientific advancement. In 1660, after Charles II was restored to the throne, science began to thrive again and the group, which included members such as Boyle and Christopher Wren, established themselves as the Royal Society. Hooke was appointed as Curator of Experiments. In this role, he set up and demonstrated experiments either of his own devising or proposed by members, which ranged from investigating the nature of air to gravity and barometric pressure. Hooke's immense skill and aptitude earned him fellowship to the Royal Society in 1663.

In this capacity, Hooke devoted himself to scientific research on a wide, diverse scale, including biology, astronomy, the nature of gravity, and optics. His abilities as an inventor and mechanic enabled him to personally carry out necessary improvements to the equipment he used, as well as to design new apparatus. Most notably, he built a compound microscope with a new screw-based mechanism to enable the viewer to focus; he also devised a way of illuminating the specimen beneath the lens. Hooke's new microscope enabled him to produce *Micrographia* in 1665—a full study of the microscopic natural world, which he observed, recorded, and illustrated in unprecedented detail. Through intricate, large-scale line drawings, Hooke revealed the structure of minute organisms, such as a flea, with an accuracy never before seen by the human eye, and in doing so produced the world's first best-selling science book. Hooke coined the term "cell," which he used in the book, and laid the foundations for the field of microbiology.

"The most **ingenious book** I ever read **in my life.**"

Samuel Pepys, 1665

Fire and fortune

After the Great Fire of London in 1666, Hooke assisted Christopher Wren in the architectural redesign of the capital. Although Wren largely took the glory for this work, Hooke designed a significant proportion